

Patient

Name: MARTIN, ELIZABETH
Date of Birth: 09/01/1955
Sex: Female
Case Number: TN24-274506
Diagnosis: Serous carcinoma, NOS

Specimen Information

Primary Tumor Site: Endometrium
Specimen Site: Uterus, NOS
Specimen ID: SX23-11637-4G
Specimen Collected: 14-Dec-2023
Test Report Date: 05-Dec-2024

Ordered By

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(859) 257-4488

Results with Therapy Associations

BIOMARKER	METHOD	ANALYTE	RESULT	THERAPY ASSOCIATION		BIOMARKER LEVEL*
Mismatch Repair Status	IHC	Protein	Proficient (Intact)	BENEFIT	pembrolizumab + lenvatinib	Level 1
MSI	Seq	DNA-Tumor	Stable			Level 2
ER	IHC	Protein	Positive 1+, 70%	BENEFIT	endocrine therapy	Level 3
PR	IHC	Protein	Positive 1+, 90%			

* Biomarker reporting classification: Level 1 – Companion diagnostic (CDx); Level 2 – Strong evidence of clinical significance or is endorsed by standard clinical guidelines; Level 3 – Potential clinical significance. Bolded benefit therapies, if present, highlight the most clinically significant findings.

Important Note

The FDA has granted regular approval of pembrolizumab in combination with lenvatinib, for the treatment of patients with advanced endometrial carcinoma that are not microsatellite instability high (MSI-H) or mismatch repair deficient (dMMR), who have disease progression following prior systemic therapy and are not candidates for curative surgery or radiation.

For information on Caris GPSai™, please refer to relevant section of the report (Additional Results Page 1).

Cancer-Type Relevant Biomarkers

Biomarker	Method	Analyte	Result
TP53	Seq	DNA-Tumor	Pathogenic Variant Exon 4 p.S94*
BRAF	Seq	DNA-Tumor	Mutation Not Detected
NTRK1/2/3	Seq	RNA-Tumor	Fusion Not Detected
RET	Seq	RNA-Tumor	Fusion Not Detected
Tumor Mutational Burden	Seq	DNA-Tumor	Low, 3 mut/Mb
ARID1A	Seq	DNA-Tumor	Mutation Not Detected

Biomarker	Method	Analyte	Result
BRCA1	Seq	DNA-Tumor	Mutation Not Detected
BRCA2	Seq	DNA-Tumor	Mutation Not Detected
CCNE1	CNA-Seq	DNA-Tumor	Intermediate
ERBB2 (Her2/Neu)	CNA-Seq	DNA-Tumor	Amplification Not Detected
	IHC	Protein	Negative Score 0
KRAS	Seq	DNA-Tumor	Mutation Not Detected

(continued on next page)

The selection of any, all, or none of the matched therapies resides solely with the discretion of the treating physician. Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, the FDA prescribing information for any therapeutic, and in accordance with the applicable standard of care. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. All trademarks and registered trademarks are the property of their respective owners.

Cancer-Type Relevant Biomarkers (continued)

Biomarker	Method	Analyte	Result	Biomarker	Method	Analyte	Result
PD-L1 (SP142)	IHC	Protein	Negative 0%	TSC1	Seq	DNA-Tumor	Mutation Not Detected
PIK3CA	Seq	DNA-Tumor	Mutation Not Detected	TSC2	Seq	DNA-Tumor	Mutation Not Detected
POLE	Seq	DNA-Tumor	Mutation Not Detected				

Genomic Signatures

Biomarker	Method	Analyte	Result
Microsatellite Instability (MSI)	Seq	DNA-Tumor	Stable
Tumor Mutational Burden (TMB)	Seq	DNA-Tumor	Result: Low 3 Low10High
Genomic Loss of Heterozygosity (LOH)	Seq	DNA-Tumor	Low - 9% of tested genomic segments exhibited LOH (assay threshold is ≥ 16%)

Genes Tested with Pathogenic or Likely Pathogenic Alterations

Gene	Method	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %
FGFR2	Seq	DNA-Tumor	Pathogenic Variant	p.S252W	7	c.755C>G	34
PPP2R1A	Seq	DNA-Tumor	Pathogenic Variant	p.P179R	5	c.536C>G	38
TP53	Seq	DNA-Tumor	Pathogenic Variant	p.S94*	4	c.281C>G	72

Unclassified alterations for DNA and RNA sequencing can be found in the MI Portal.

Formal nucleotide nomenclature and gene reference sequences can be found in the Appendix of this report.

Variants of Uncertain Significance can be found in the MI Portal.

Human Leukocyte Antigen (HLA) Genotype Results

The impact of HLA genotypes on drug response and prognosis is an active area of research. These results can help direct patients to clinical trials recruiting for specific genotypes. Please see www.clinicaltrials.gov for more information.

Gene	Method	Analyte	Genotype
MHC CLASS I			
HLA-A	Seq	DNA-Tumor	A*23:01, A*29:02
HLA-B	Seq	DNA-Tumor	B*44:03, B*44:03
HLA-C	Seq	DNA-Tumor	C*04:01, C*16:01

HLA genotypes with only one allele are either homozygous or have loss-of-heterozygosity at that position.

Immunohistochemistry Results

Biomarker	Result	Biomarker	Result
ER	Positive 1+, 70%	MSH6	Intact nuclear expression
ERBB2 (Her2/Neu)	Negative Score 0	PD-L1 (SP142)	Negative 0%
Mismatch Repair Status	Proficient (Intact)	PMS2	Intact nuclear expression
MLH1	Intact nuclear expression	PR	Positive 1+, 90%
MSH2	Intact nuclear expression		

Genes Tested with Indeterminate Results by Tumor DNA Sequencing

SMARCA2																			
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Genes in this table were ruled indeterminate due to low coverage for some or all exons.

Genes Tested with Intermediate CNA Results by Tumor DNA Sequencing

CCNE1	ERBB3																		
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The results in this report were curated to represent biomarkers most relevant for the submitted cancer type. These include results important for therapeutic decision-making, as well as notable alterations in other biomarkers known to be involved in oncogenesis. Additional results, including genes with normal findings, variants of uncertain significance, or unclassified alterations can be found in the MI Portal at miportal.carismolecularintelligence.com. If you do not have an MI Portal account, or need assistance accessing it, please contact Caris Customer Support at (888) 979-8669.

Notes of Significance

SEE APPENDIX FOR DETAILS

Clinical Trials Connector™ opportunities based on biomarker expression: 371 Targeted Therapies. See page 5 for details.

Specimen Information

Specimen ID: SX23-11637-4G

Specimen Collected: 12/14/2023

Specimen Received: 11/27/2024

Testing Initiated: 11/29/2024

Test Ordered*: MI Profile™ (MI Tumor Seek Hybrid™ + IHCs and Other Tests by Tumor Type)

* If the submitted specimen is inadequate, only a subset of the ordered testing may be reported.

Gross Description: 1 (A) Paraffin Block - Client ID(SX23-11637-4G) from Baptist Health - Lexington, Lexington, KY, with the corresponding pathology report labeled "SX23-11637".

Dissection Information: Molecular testing of this specimen was performed after harvesting of targeted tissues with an approved manual microdissection technique. Candidate slides were examined under a microscope and areas containing tumor cells (and separately normal cells, when necessary for testing) were circled. A laboratory technician harvested targeted tissues for extraction from the marked areas using a dissection microscope.

Clinical Trials Connector™

The Clinical Trials Connector lists agents that are matched to available clinical trials according to biomarker status. In some instances, older-generation agents may still be relevant in the context of new combination strategies and, therefore, will still appear on this report.

Therapeutic agents listed below may or may not be currently FDA approved for the tumor type tested.

Please see <https://clinicaltrials.gov/> for more information.

TARGETED THERAPIES (371)				
Drug Class	Biomarker	Method	Analyte	Investigational Agent(s)
Anti-hormonal therapy (115)	ER	IHC	Protein	anastrozole, exemestane, fulvestrant, goserelin, letrozole, leuprolide, tamoxifen
	PR	IHC	Protein	
FGFR-targeted therapy (256)	FGFR2	NGS	DNA-Tumor	AZD4547, pemigatinib, erdafitinib, futibatinib, bemarituzumab, 3D185, derazantinib, ponatinib, regorafenib, TT-00420, anlotinib, ICP-192, lenvatinib, nintedanib

() = represents the total number of clinical trials identified by the Clinical Trials Connector for the provided drug class or table.

The Clinical Trials Connector may include trials that enroll patients with additional screening of molecular alterations. In some instances, only specific gene variants may be eligible.

Disclaimer

Decisions on patient care and treatment must be based on the independent medical judgment of the treating physician, taking into consideration all available information concerning the patient's condition, prescribing information for any therapeutic, and in accordance with the applicable standard of care. Drug associations provided in this report do not guarantee that any particular agent will be effective for the treatment of any patient or for any particular condition. Caris Life Sciences[®] expressly disclaims and makes no representation or warranty whatsoever relating, directly or indirectly, to the performance of services, including any information provided and/or conclusions drawn from therapies that are included or omitted from this report. Whether or not a particular patient will benefit from a selected therapy is based on many factors and can vary significantly. The selection of therapy, if any, resides solely in the discretion of the treating physician and the tests should not be considered a companion diagnostic.

Caris MPI, Inc. d/b/a Caris Life Sciences is certified under the Clinical Laboratory Improvement Amendments (CLIA) as qualified to perform high complexity clinical laboratory testing, including all Caris molecular profiling assays. Individual assays that are available through Caris molecular profiling include both Laboratory Developed Tests (LDT) and U.S. Food and Drug Administration (FDA) approved or cleared tests. In addition, certain tests have been CE-marked as a general IVD under the In Vitro Diagnostic Directive (IVDD) 98/79/EC. Offered LDTs were developed and their performance characteristics determined by Caris. Certain tests have not been cleared or approved by the FDA. Caris LDTs are used for clinical purposes. They are not investigational or for research. Caris' CLIA certification number is located at the bottom of each page of this report.

The information presented in the Clinical Trials Connector™ section of this report, if applicable, is compiled from sources believed to be reliable and current. However, the accuracy and completeness of the information provided herein cannot be guaranteed. The clinical trials information present in the biomarker description was compiled from www.clinicaltrials.gov. The contents are to be used only as a guide, and health care providers should employ their best comprehensive judgment in interpreting this information for a particular patient. Specific eligibility criteria for each clinical trial should be reviewed as additional inclusion criteria may apply.

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Caris molecular testing is subject to Caris' intellectual property. Patent www.CarisLifeSciences.com/ip.

Professional Component Performed: -- Location 1, Chandler, AZ. Jose de los Santos, MD, Lab Director.



Electronic Signature

Jose de los Santos, MD

05-Dec-2024



The Caris GPSai (Caris Genomic Probability Score – Artificial Intelligence) is a cancer-type similarity assessment which compares the characteristics of a patient's tumor against other tumors in the Caris database. Caris GPSai analyzes a tumor's molecular signature and provides the prevalence of that signature in the Caris Life Sciences genomic and transcriptomic database across 90 distinct cancer categories.

Most Likely Tumor Type: Uterine Serous Carcinoma/Uterine Papillary Serous Carcinoma

97%

The predicted probability of each cancer type that can be detected by the product are shown below. The total probabilities of the top level cancer type (blue) total 100%. The subtypes (indented) sum to the total probability predicted for the top level cancer type. Discrepancies between expected sums and displayed sums may be exhibited due to rounding.

0	Adrenal Cortical Carcinoma
0	Bladder/Urinary Tract
0	Urothelial Carcinoma
0	Bladder Adenocarcinoma
0	Bowel
0	Appendiceal Adenocarcinoma
0	Colorectal Adenocarcinoma
0	Small Bowel Carcinoma
0	Breast
0	Metaplastic Breast Cancer
0	Breast Invasive Lobular Carcinoma
0	Breast Invasive Ductal Carcinoma
0	CNS/Brain
0	Diffuse Glioma
0	Meningioma
99	Cervix/Uterine Carcinoma
97	Uterine Serous Carcinoma/Uterine Papillary Serous Carcinoma
1	Uterine Carcinosarcoma/Uterine Malignant Mixed Mullerian Tumor
0	Uterine Clear Cell Carcinoma
0	Cervical Adenocarcinoma
2	Uterine Endometrioid Carcinoma
0	Cutaneous Squamous Cell Carcinoma
0	Esophagus/Stomach
0	Esophageal Squamous Cell Carcinoma
0	Esophagogastric Adenocarcinoma
0	Stomach Adenocarcinoma
0	Adenocarcinoma of the Gastroesophageal Junction
0	Esophageal Adenocarcinoma
0	Germ Cell Tumor
0	Hematological
0	Hepatocellular Carcinoma
0	Kidney
0	Renal Cell Carcinoma
0	Chromophobe Renal Cell Carcinoma
0	Papillary Renal Cell Carcinoma
0	Renal Clear Cell Carcinoma
0	Wilms Tumor
0	Melanoma
0	Mesothelioma
0	Neuroendocrine Neoplasm
0	Well/Moderately-Differentiated Neuroendocrine Tumor
0	Poorly-Differentiated Neuroendocrine Carcinoma
0	Large Cell Neuroendocrine Carcinoma
0	Small Cell Neuroendocrine Carcinoma
0	Paraganglioma/Pheochromocytoma
0	Merkel Cell Carcinoma

0	Non-Small Cell Lung Carcinoma
0	Lung Squamous Cell Carcinoma
0	Lung Adenocarcinoma
0	Orogenital Squamous Cell Carcinoma
0	Ovarian Epithelial Tumor
0	Ovarian Carcinosarcoma/Malignant Mixed Mesodermal Tumor
0	Serous Ovarian/Fallopian Tube/Peritoneal
0	Low-Grade Serous Ovarian/Fallopian Tube/Peritoneal Cancer
0	High-Grade Serous Ovarian/Fallopian Tube/Peritoneal Cancer
0	Mucinous Ovarian Cancer
0	Clear Cell Ovarian Cancer
0	Endometrioid Ovarian Cancer
0	Pancreatobiliary
0	Gallbladder Cancer
0	Pancreatic Adenocarcinoma
0	Cholangiocarcinoma
0	Peripheral Nervous System
0	Schwannoma
0	Neuroblastoma
0	Malignant Peripheral Nerve Sheath Tumor
0	Prostate Adenocarcinoma
0	Salivary Gland Tumor
0	Sex Cord Stromal Tumor
0	Granulosa Cell Tumor
0	Sertoli-Leydig Cell Tumor
0	Soft Tissue/Bone
0	Gastrointestinal Stromal Tumor
0	Leiomyosarcoma
0	Angiosarcoma
0	Rhabdomyosarcoma
0	Synovial Sarcoma
0	Osteosarcoma
0	Liposarcoma
0	Chondrosarcoma
0	Endometrial Stromal Sarcoma
0	Ewing Sarcoma
0	Thymic Carcinoma
0	Thyroid
0	Medullary Thyroid Cancer
0	Hurthle Cell Thyroid Cancer
0	Well-Differentiated Thyroid Cancer
0	Follicular Thyroid Cancer
0	Papillary Thyroid Cancer
0	Anaplastic Thyroid Cancer

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TN24-274506

PHYSICIAN: TRICIA L. FREDERICKS

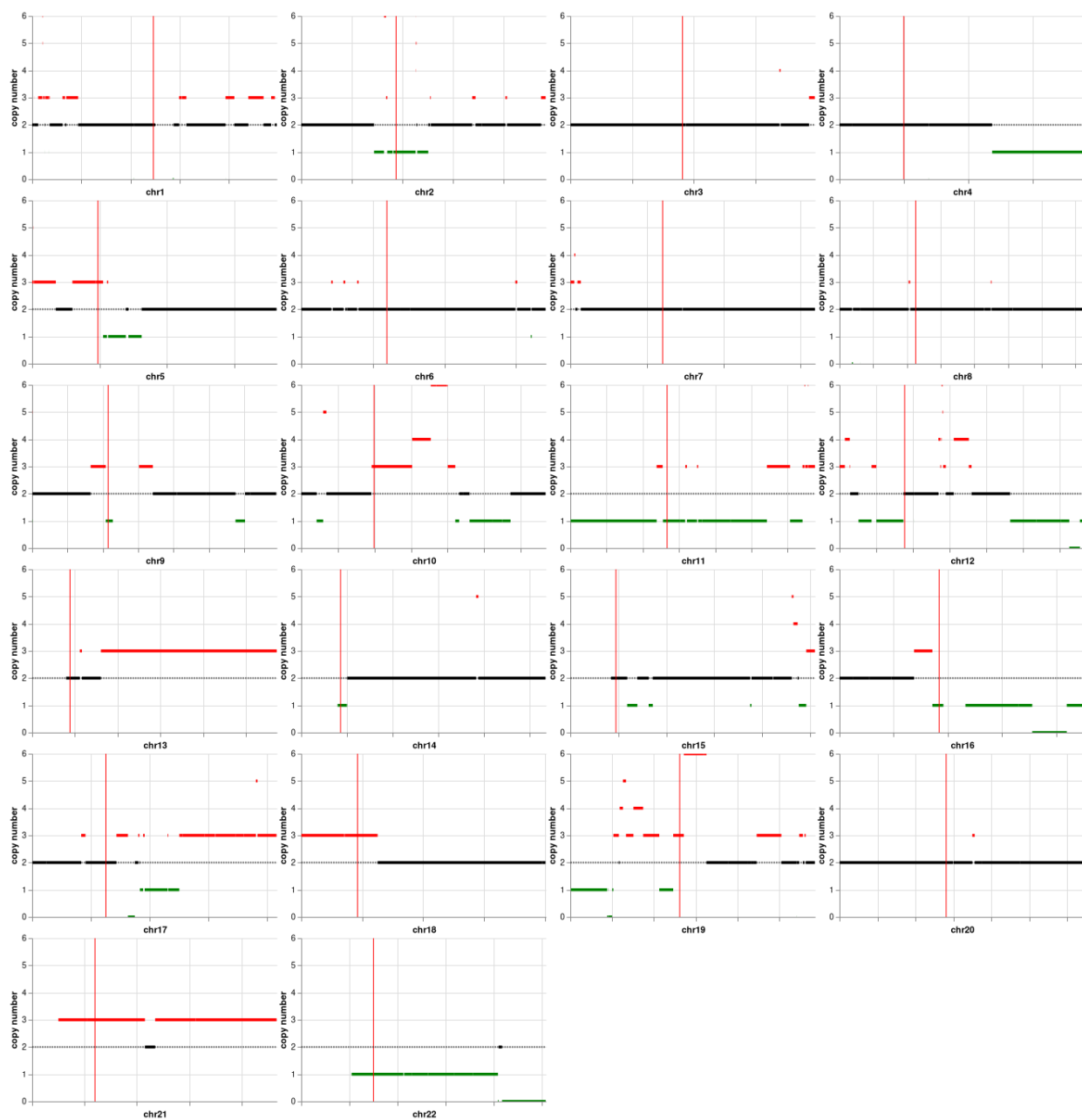
Gene Expression

Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type	Gene	Percentile in Cancer Type
ADORA2A	81	FGFR2	96	NTRK2	88
ALK	84	FGFR3	62	NTRK3	96
ARID1A	78	FOLR1	74	PDCD1	97
ATM	20	HRAS	42	PDCD1LG2	83
AURKA	20	IGF1R	50	PGR	89
BRAF	76	KDM1A	72	PIK3CA	16
BRCA1	26	KDR	76	PRAME	40
BRCA2	74	KRAS	2	PTEN	40
BRD4	82	LAG3	95	RB1	68
CCND1	33	MAGEA4	76	RET	90
CCNE1	75	MDM2	91	ROR1	81
CD274	39	MET	41	ROR2	73
CD276	5	MKI67	70	ROS1	90
CDKN2A	39	MSLN	72	SRC	34
CEACAM5	78	MTAP	20	TACSTD2	22
CLDN18	78	MTOR	40	TGFB1	32
CLDN6	68	MUC1	45	TOP1	79
CTLA4	80	MUC16	41	TP53	22
CTNNB1	30	MYC	61	TSC1	43
EGFR	50	NECTIN4	86	TSC2	92
EPHA2	67	NF1	46	VEGFA	8
ERBB2	26	NRAS	23	XPO1	84
ERBB3	100	NRG1	84		
ESR1	80	NTRK1	94		

Gene Expression of Selected Genes by Whole Transcriptome Sequencing (WTS) Methods:

Gene expression is derived from whole transcriptome sequencing. Relative expression of genes are calculated as normalized values using Transcripts per Million Molecules or TPM. TPM is presented as a percentile derived by comparison to a distribution of Caris' internal cohort of the tumor-type profiled. Selected genes reported in this section were chosen based on their tumor-type specific relevance for matching to clinical trials, or tumor type subclassification.

Karyotype



Karyotyping using Copy Number Analysis by Whole Exome Sequencing (WES) Methods:

Whole exome sequencing in combination with interrogation of single nucleotide polymorphisms (SNPs) tiled throughout the genome, allows for the identification and visualization of cytogenetic aberrations.

Somatic structural variants like whole or partial chromosome duplications or deletions, are important for cancer development and progression, and may identify clinically actionable alterations.

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Mutational Analysis by Next-Generation Sequencing (NGS)

TUMOR MUTATIONAL BURDEN	
Mutations / Megabase	Result
3	Low

TMB

Tumor Mutational Burden (TMB) is defined as the number of somatic non-synonymous mutations per million bases of sequenced DNA in a tumor sample. Tumors with high TMB may increase the number of neoantigens which is hypothesized to increase T-cell reactivity and potential for response to immune checkpoint inhibitors. TMB analysis was performed based on next generation sequencing analysis of genomic DNA isolated from a tumor sample.

MICROSATELLITE INSTABILITY ANALYSIS	
Test	Result
MSI	Stable

MSI

Microsatellite instability (MSI) status is a measure of the number of somatic mutations within short, repeated sequences of DNA (microsatellites). MSI-High status can indicate that the tumor has a defect in mismatch repair (MMR) abrogating the ability to correct mistakes during DNA replication. Tumors with MSI-high status may increase the number of neoantigens which is hypothesized to increase T-cell reactivity and potential for response to immune checkpoint inhibitors. Tumor-only microsatellite instability status by NGS (MSI-NGS) is measured by the direct analysis of known microsatellite regions sequenced in the CMI NGS panel.

GENOMIC LOSS OF HETEROZYGOSITY	
Test	Result
Genomic Loss of Heterozygosity (LOH)	Low - 9% of tested genomic segments exhibited LOH (assay threshold is $\geq 16\%$)

LOH

To calculate genomic loss-of-heterozygosity (LOH), the 22 autosomal chromosomes are split into 552 segments and the LOH of single nucleotide polymorphisms (SNPs) within each segment is calculated. Caris WES data consist of approximately 250k SNPs spread across the genome. SNP alleles with frequencies skewed towards 0 or 100% indicate LOH (heterozygous SNP alleles have a frequency of 50%). The final call of genomic LOH is based on the percentage of all 552 segments with observed LOH.

Additional Next-Generation Sequencing results continued on the next page. >

PATIENT: MARTIN, ELIZABETH (01-SEP-1955)

TN24-274506

PHYSICIAN: TRICIA LEE, FREDERICKS

Mutational Analysis by Next-Generation Sequencing (NGS)

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
FGFR2	DNA-Tumor	Pathogenic Variant	p.S252W	7	c.755C>G	34	NM_000141.4

Interpretation: A common pathogenic activating mutation, p.S252W, was detected in FGFR2. In the germline, p.S252W is causal for Apert syndrome.

FGFR2 is a receptor for fibroblast growth factor. Activation of FGFR2 through mutation and amplification has been noted in a number of cancers. Somatic mutations of the fibroblast growth factor receptor 2 (FGFR2) tyrosine kinase are present in endometrial carcinoma, lung squamous cell carcinoma, cervical carcinoma, and melanoma. In the endometrioid histology of endometrial cancer, the frequency of FGFR2 mutation is 16% and the mutation is associated with shorter disease free survival in patients diagnosed with early stage disease. Loss of function FGFR2 mutations occur in about 8% melanomas and contribute to melanoma pathogenesis. Germline mutations in FGFR2 are associated with numerous medical conditions that include congenital craniofacial malformation disorders, Apert syndrome and the related Pfeiffer and Crouzon syndromes.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
PPP2R1A	DNA-Tumor	Pathogenic Variant	p.P179R	5	c.536C>G	38	NM_014225.5

Interpretation: A pathogenic mutation was detected in PPP2R1A (PMID: 21435433).

This gene encodes a constant regulatory subunit of protein phosphatase 2. Protein phosphatase 2 is one of the four major Ser/Thr phosphatases, and it is implicated in the negative control of cell growth and division. It consists of a common heteromeric core enzyme, which is composed of a catalytic subunit and a constant regulatory subunit, that associates with a variety of regulatory subunits.

Gene	Analyte	Variant Interpretation	Protein Alteration	Exon	DNA Alteration	Variant Frequency %	Transcript ID
TP53	DNA-Tumor	Pathogenic Variant	p.S94*	4	c.281C>G	72	NM_000546.5

Interpretation: A pathogenic mutation was detected in TP53.

TP53, or p53, plays a central role in modulating response to cellular stress through transcriptional regulation of genes involved in cell-cycle arrest, DNA repair, apoptosis, and senescence. Inactivation of the p53 pathway is essential for the formation of the majority of human tumors. Mutation in p53 (TP53) remains one of the most commonly described genetic events in human neoplasia, estimated to occur in 30-50% of all cancers. Generally, presence of a disruptive p53 mutation is associated with a poor prognosis in all types of cancers, and diminished sensitivity to radiation and chemotherapy. Germline p53 mutations are associated with the Li-Fraumeni syndrome (LFS) which may lead to early-onset of several forms of cancer currently known to occur in the syndrome, including sarcomas of the bone and soft tissues, carcinomas of the breast and adrenal cortex (hereditary adrenocortical carcinoma), brain tumors and acute leukemias.

Additional Next-Generation Sequencing results continued on the next page. >

PATIENT: MARTIN, ELIZABETH (01-SEP-1955)

TN24-274506

PHYSICIAN: TRICIA LEE, FREDERICKS

Mutational Analysis by Next-Generation Sequencing (NGS)

GENES TESTED WITH INDETERMINATE* RESULTS BY TUMOR DNA SEQUENCING					
SMARCA2					

* Genes in this table were ruled indeterminate due to low coverage for some or all exons.

For a complete list of genes tested, visit www.CarisMolecularIntelligence.com/profilemenu.

NGS Methods

Direct sequence analysis was performed on genomic DNA isolated from a micro-dissected tumor sample using Illumina NovaSeq 6000 sequencers. A hybrid pull-down panel of baits was used to enrich more than 700 clinically relevant genes along with > 20,000 other genes. Sequence data is analyzed using a customized bioinformatics pipeline to detect sequencing variants, copy number alterations (amplifications and deletions) indels and HLA genotypes. In addition, genomic signatures for tumor mutational burden (TMB), microsatellite instability (MSI), genomic loss-of-heterozygosity (LOH) or HRD-Genomic Scar Score (HRD-GSS), and homologous recombination deficiency (HRD) are reported when applicable. For a complete list of what is covered by the assay, and genes with partial coverage, please contact Caris Customer Support.

Copy Number Alterations by Next-Generation Sequencing (NGS)

GENES TESTED WITH INTERMEDIATE CNA RESULTS					
CCNE1	ERBB3				

CNA Methods

The copy number alteration (CNA) of each exon is determined by a calculation using the average sequencing depth of the sample along with the sequencing depth of each exon and comparing this calculated result to a pre-calibrated value. A complete list of genes for reporting copy number alterations, including amplifications and deletions, is available upon request.

Gene Fusion and Transcript Variant Detection by RNA Sequencing

Whole Transcriptome Sequencing (WTS) Methods

Gene fusion and variant transcript detection were performed on RNA isolated from a tumor sample using next generation sequencing. The assay also detects fusions occurring at known and novel breakpoints within genes. The genes included in this report represent the subset of genes associated with cancer. The complete list of unclassified alterations is available by request.

Protein Expression by Immunohistochemistry (IHC)

Biomarker	Patient Tumor			Thresholds
	Staining Intensity (0, 1+, 2+, 3+)	Percent of cells	Result	Conditions for a Positive Result:
ER	1 +	70	Positive	Intensity $\geq 1+$ and $\geq 10\%$ of cells stained
ERBB2 (Her2/Neu)	0	100	Negative	Intensity $\geq 3+$ and $\geq 10\%$ of cells stained
PD-L1 (SP142)	0	100	Negative	Intensity $\geq 2+$ and $\geq 5\%$ of cells stained
PR	1 +	90	Positive	Intensity $\geq 1+$ and $\geq 10\%$ of cells stained

MISMATCH REPAIR (MMR) PROTEINS	
Biomarker	Result
MLH1	Intact nuclear expression
MSH2	Intact nuclear expression
MSH6	Intact nuclear expression
PMS2	Intact nuclear expression
Mismatch Repair Protein Status	Proficient (Intact)

Mismatch Repair Protein Status result interpretation: Proficient (Intact) – No evidence of deficient mismatch repair (no loss of nuclear expression of any MMR protein); Deficient (Loss) – Loss of nuclear expression of one or more MMR proteins

Clones used: ER (SP1), ERBB2 (Her2/Neu) (4B5), MLH1 (M1), MSH2 (G219-1129), MSH6 (SP93), PD-L1 (SP142), PMS2 (A16-4), PR (1E2).



Electronic Signature
Jose de los Santos, MD
12/04/2024

IHC Methods

The Laboratory Developed Tests (LDT) immunohistochemistry (IHC) assays were developed and their performance characteristics determined by Caris Life Sciences. These tests have not been cleared or approved by the US Food and Drug Administration. The FDA has determined that such clearance or approval is not currently necessary. Interpretations of all immunohistochemistry (IHC) assays were performed manually or with the assistance of an AI-based image analysis tool by a board certified pathologist using a microscope and/or digital whole slide image(s).

The following IHC assays were performed using FDA-approved companion diagnostic or FDA-cleared tests consistent with the manufacturer's instructions: ALK (VENTANA ALK (D5F3) CDx Assay, Ventana), ER (CONFIRM anti-Estrogen Receptor (ER) (SP1), Ventana), FOLR1 (VENTANA FOLR1-2.1 RxDx, Ventana), CLDN18 (VENTANA, 43-14A RxDx Assay), PR (CONFIRM anti-Progesterone Receptor (PR) (1E2), Ventana), HER2/neu (PATHWAY anti-HER-2/neu (4B5), Ventana), Ki-67 (MIB-1 pharmaDx, Dako), PD-L1 22c3 (pharmDx, Dako), PD-L1 SP142 (VENTANA, non-small cell lung cancer), PD-L1 28-8 (pharmDx, Dako, gastric / GEJ, non-small cell lung cancer), PD-L1 SP263 (Ventana, non-small cell lung cancer), and Mismatch Repair (MMR) proteins (MLH1, MSH2, MSH6, and PMS2; VENTANA MMR RxDx Panel, Ventana).

HER2 results and interpretation follow the ASCO/CAP scoring criteria. Bartley, A.N., J.A. Ajani, et al. (2016). "HER2 testing and clinical decision making in gastroesophageal adenocarcinoma: guideline from the College of American Pathologists, American Society for Clinical Pathology, and the American Society of Clinical Oncology". J Clin Oncol. 35(4):446-464.

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References

#	Drug	Biomarker	Reference
1	endocrine therapy	ER, PR	Singh, M., K.K., Leslie, et al. (2007) "Relationship of estrogen and progesterone receptors to clinical outcome in metastatic endometrial carcinoma: a Gynecologic Oncology Group Study." Gynecol Oncol. 2007 Aug;106(2):325-33. View Citation Online
2	pembrolizumab + lenvatinib	MSI, Mismatch Repair Status	Makker, V., M. Taylor, et al. (2019). "Lenvatinib plus pembrolizumab in patients with advanced endometrial cancer: an interim analysis of a multicentre, open-label, single-arm, phase 2 trial." Lancet Oncol, 20:711-18. View Citation Online